UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

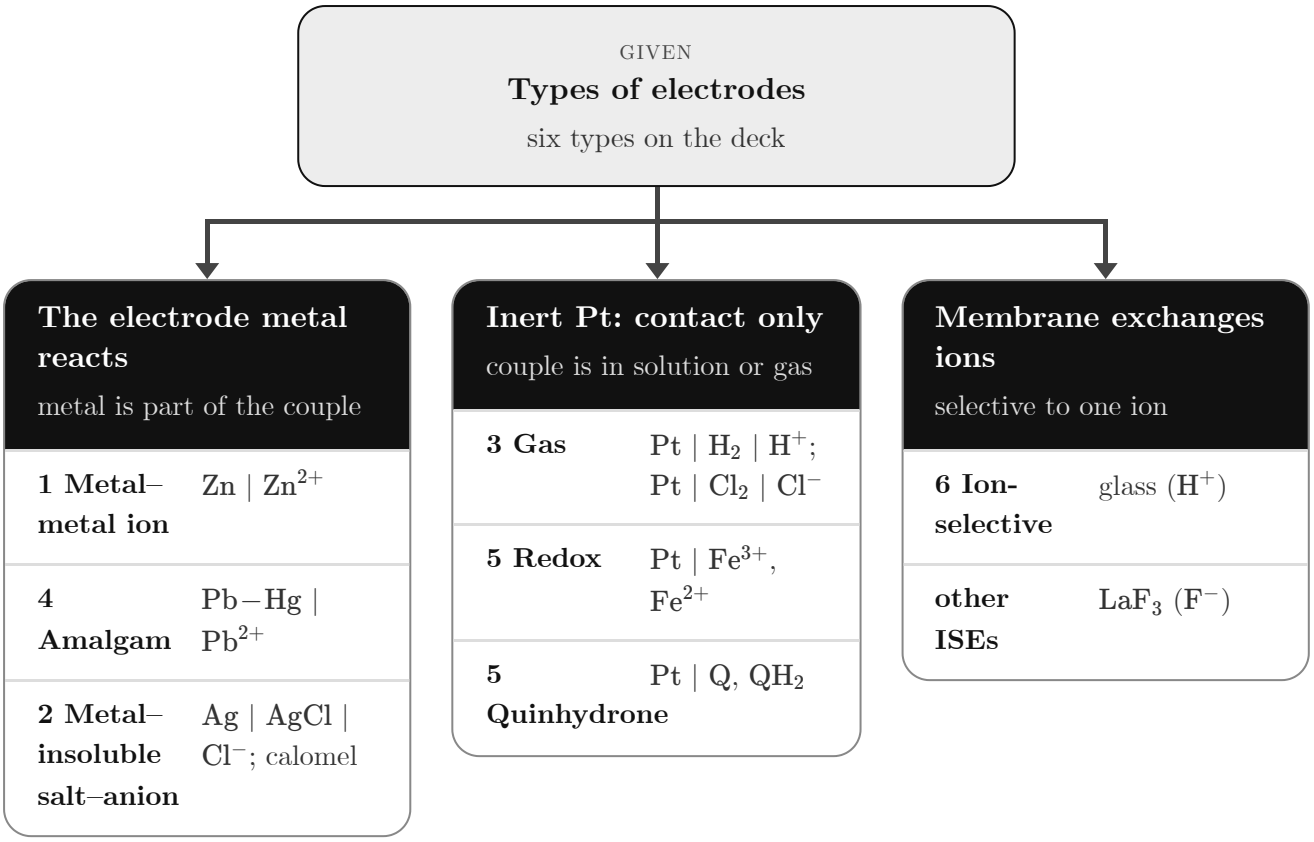
Class 003 · Types of electrodes · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·5	Deck 003, slide 5 (rendered .pptx)
T003 00:07:50	AV-summary transcript 003, segment start
PDF p.57	Older consolidated Unit 2 slide PDF (8enterprise), page 57
ESA Dec-23 2d	Past paper: ESA Dec 2023, question 2.d
ESA Jan-25 s1 2b	Past paper: ESA Jan 2025 set 1, question 2.b

Contents

1. Figure F1.1 · Classification of electrodes (the six types in three groups).
2. Figure F1.2 · Writing the Nernst equation for any electrode.
3. Figure F1.3 · Which type is this electrode? (from its representation)
4. Figure F6.1 · Potential of a quinhydrone electrode from pH.
5. The six types of electrodes at a glance
6. Gas electrode vs redox electrode (ESA Dec 2023 note)
7. Three electrodes whose potential depends on pH

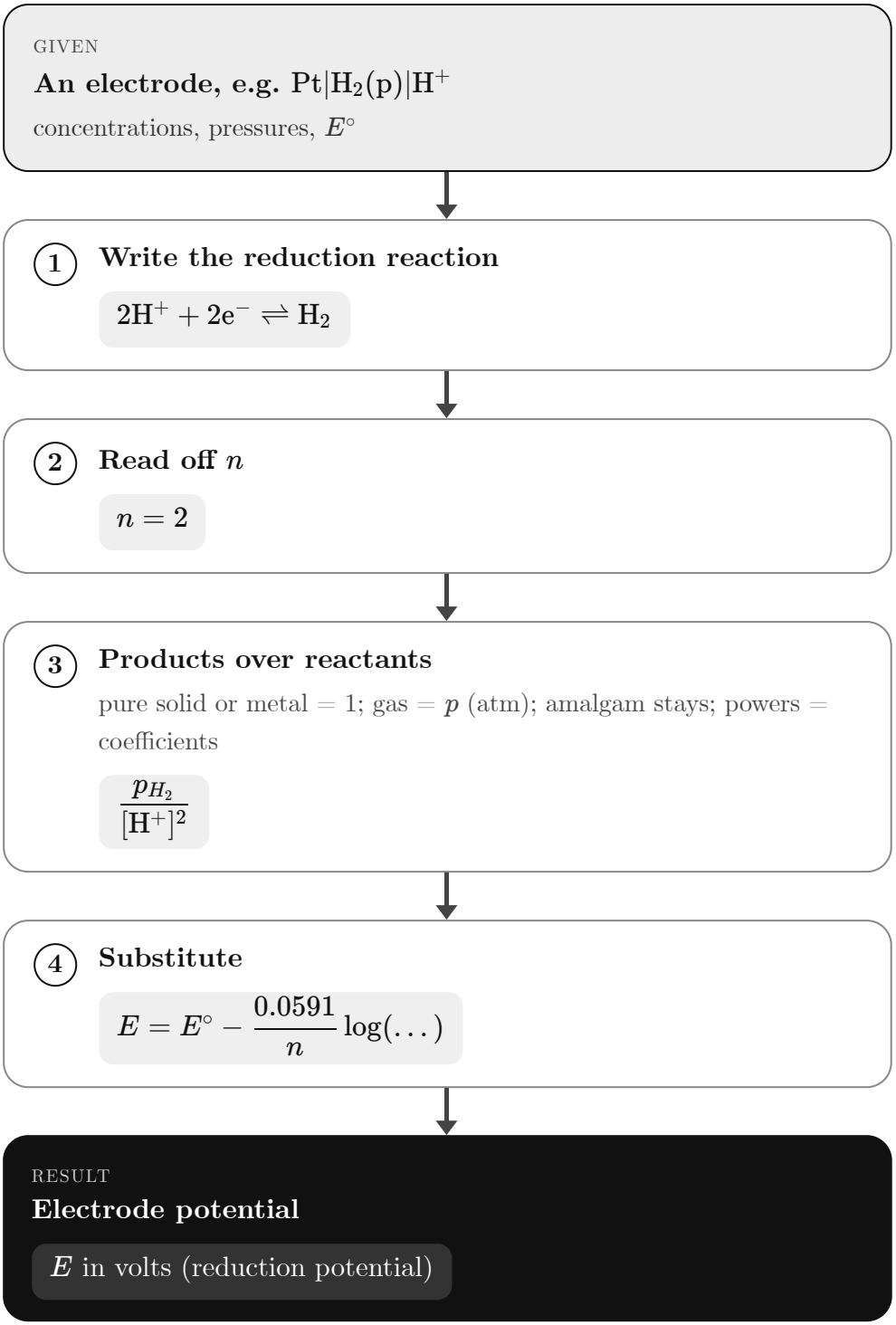
1 Types of electrodes: the map



Remark. The three groups are added for clarity; the numbers are the deck's. The quinhydrone electrode is a special case of type 5.

Figure F1.1. Classification of electrodes (the six types in three groups). ESA ★★

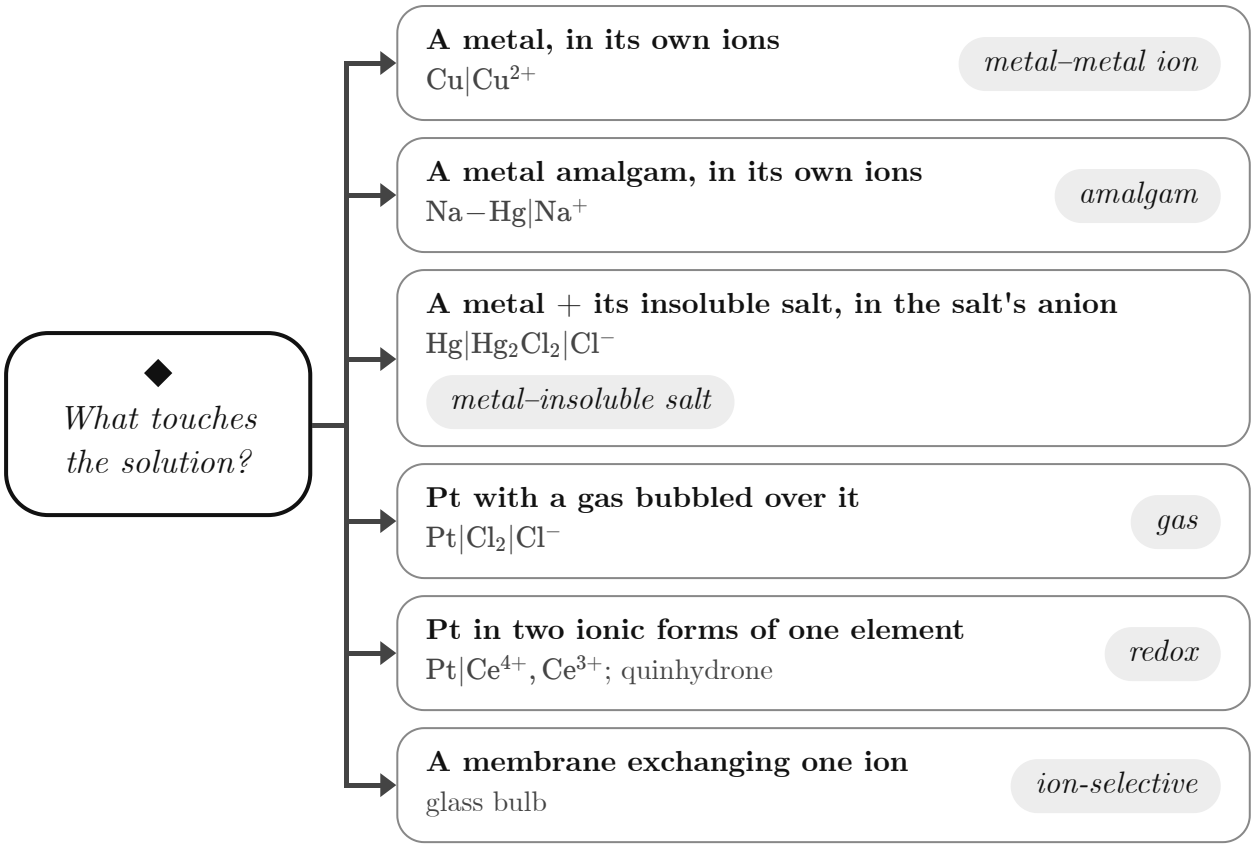
ADDED FOR CLARITY S003·3 · S003·4 · S003·5 · S003·6 · S003·7 · S003·8 · S003·9 · T003 00:00:16



Exam trap. Writing the oxidation instead of the reduction flips the log term and gives the wrong sign.

Figure F1.2. Writing the Nernst equation for any electrode. ESA ★★

T003 00:01:38 · T003 00:07:50 · T003 00:11:06



Remark. Pt never takes part in the reaction; if Pt appears, the electrode is gas or redox.

Figure F1.3. Which type is this electrode? (from its representation) ESA ★

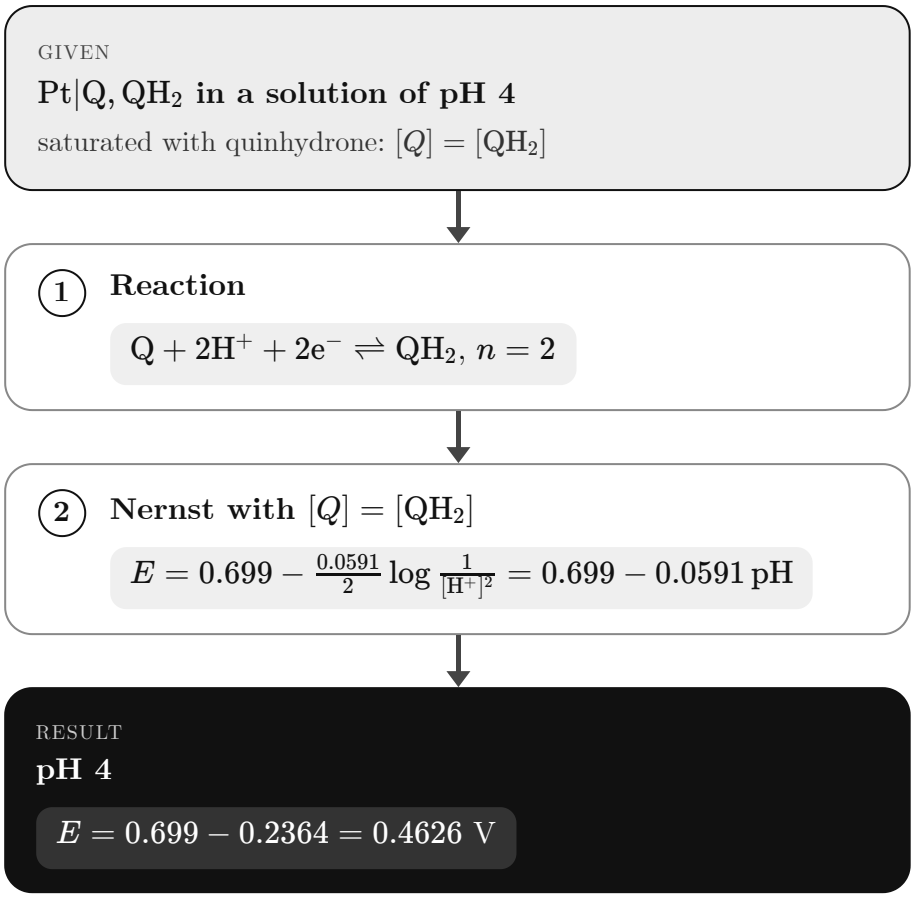
S003·3 · S003·4 · S003·5 · S003·6 · S003·7 · S003·9

The six types of electrodes at a glance

Type (responds to)	Representation	Electrode reaction	Nernst equation, 298 K
1 Metal–metal ion *(cation)*	$\mathrm{Zn}\vert\mathrm{Zn^{2+}}$	$\mathrm{M^{n+}} + \mathrm{ne^-} \rightleftharpoons \mathrm{M}$	$E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$
2 Metal–insoluble salt–anion *(anion)*	$\mathrm{Ag}\vert\mathrm{AgCl(s)}\vert\mathrm{Cl^-}$	$\mathrm{AgCl} + \mathrm{e^-} \rightleftharpoons \mathrm{Ag} + \mathrm{Cl^-}$	$E^\circ - 0.0591\log[\mathrm{Cl^-}]$
3 Gas *($\mathrm{H^+}$, p)*	$\mathrm{Pt}\vert\mathrm{H_2}\vert\mathrm{H^+}$	$2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{H_2}$	$E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2}$
4 Amalgam *(cation, amalgam)*	$\mathrm{Pb-Hg}\vert\mathrm{Pb^{2+}}$	$\mathrm{Pb^{2+}} + 2\mathrm{e^-} \rightleftharpoons \mathrm{Pb(Hg)}$	$E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Pb\text{-}Hg}]}{[\mathrm{Pb^{2+}}]}$
5 Redox *(ratio red/ox)*	$\mathrm{Pt}\vert\mathrm{Sn^{4+}}, \mathrm{Sn^{2+}}$	$\mathrm{Sn^{4+}} + 2\mathrm{e^-} \rightleftharpoons \mathrm{Sn^{2+}}$	$E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$
5 Quinhydrone *(pH)*	$\mathrm{Pt}\vert\mathrm{Q}, \mathrm{QH_2}$	$\mathrm{Q} + 2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{QH_2}$	$0.699 - \frac{0.0591}{2}\log\frac{[\mathrm{QH_2}]}{[Q][\mathrm{H^+}]^2}$
6 Ion-selective (glass) *(one ion, $\mathrm{H^+}$)*	glass membrane	ion exchange at the membrane	$E^\circ_G + 0.0591\log[\mathrm{H^+}]$

S003·3 · S003·4 · S003·5 · S003·6 · S003·7 · S003·8 · S003·9

6 Oxidation–reduction electrodes and the quinhydrone electrode



Remark. Run it backwards to measure pH: $\mathrm{pH} = (0.699 - E)/0.0591$ (valid below about pH 8).

Figure F6.1. Potential of a quinhydrone electrode from pH. ESA ★★

S003·8

Gas electrode vs redox electrode (ESA Dec 2023 note)

	Gas electrode	Oxidation–reduction electrode
Construction	Gas bubbled over inert Pt foil in a solution of its ions	Inert Pt in a solution of both oxidised and reduced forms
Role of Pt	Contact; helps set up gas–ion equilibrium	Contact only
Example	$\mathrm{Pt}\vert\mathrm{H_2}\vert\mathrm{H^+}$, $\mathrm{Pt}\vert\mathrm{Cl_2}\vert\mathrm{Cl^-}$	$\mathrm{Pt}\vert\mathrm{Fe^{3+}},\mathrm{Fe^{2+}}$, $\mathrm{Pt}\vert\mathrm{Sn^{4+}},\mathrm{Sn^{2+}}$
Reaction	$2\mathrm{H^+} + 2\mathrm{e^-} \rightleftharpoons \mathrm{H_2}$	$\mathrm{Sn^{4+}} + 2\mathrm{e^-} \rightleftharpoons \mathrm{Sn^{2+}}$
Nernst	$E^\circ - \frac{0.0591}{2}\log\frac{p_{H_2}}{[\mathrm{H^+}]^2}$	$E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$
Potential set by	$[\mathrm{H^+}]$ and gas pressure	ratio [reduced]/[oxidised]

S003·5 · S003·7 · ESA Dec-23 2d

7 Ion-selective (membrane) electrode

Three electrodes whose potential depends on pH

	Hydrogen	Quinhydrone	Glass
Type	gas	redox (special case)	ion-selective
E at 298 K	$-0.0591\,\mathrm{pH}$ (1 atm)	$0.699 - 0.0591\,\mathrm{pH}$	$E^\circ_G - 0.0591\,\mathrm{pH}$
Needs	H_2 gas, platinised Pt	quinhydrone, Pt	glass bulb + reference
Limit	poisoned easily; awkward gas supply	not above pH 8	calibrate E°_G with a buffer

ADDED FOR CLARITY S003·5 · S003·8 · S003·9